

Hanwen Zhang

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EDUCATION

Sichuan University

Sep 2024 – Expected 2028

B.Eng. in Computer Science and Technology; Honors Class

Chengdu, China

- **GPA: 3.96/4.00**, ranked **1/12** in the Honors Class.
- Relevant coursework: Theory of Optimization (98), Probability and Statistics (97), Introduction to Artificial Intelligence (96), Introduction to Multimodal Learning (90).

RESEARCH INTERESTS

Spatial reasoning in multimodal models and generative world models for embodied agents, building on a background in generative 3D vision and video diffusion.

RESEARCH EXPERIENCE

City University of Hong Kong

Jul 2026 – Present

Research Intern, Department of Data Science (Supervisor: Xinke Li)

Hong Kong & Remote

- Investigating sparse-view 3D scene reconstruction with video diffusion, predicting unobserved views from a small set of input images.
- Reproduced the GenRecon and ReconViaGen pipelines and analyzed where reconstruction errors arise across their stages.

Bone-to-Shape: Single-Image Animal Reconstruction

Jan 2026 – Sep 2026

Project Lead, Provincial Undergraduate Innovation Training Program

Sichuan University

- Built a single-image animal reconstruction pipeline combining AniMer pose and shape priors, video-diffusion novel-view synthesis, and 3D Gaussian Splatting (3DGS).
- Investigated how anatomical and body-shape priors constrain geometry across generated views; this work led to the patent application listed below.

Deep-Hole Guardian: Robotic Deep-Hole Defect Inspection

Jan 2026 – Sep 2026

Team Member, University-Level Undergraduate Innovation Training Program

Sichuan University

- Processed depth-camera point clouds for deep blind-hole defect detection in a SCARA-robot inspection system.

PUBLICATIONS & PATENTS

Yijia Chen, Ze Zhang, **Hanwen Zhang**, Jiaqi Yang, Bingye Zhou, and Zhong Yuan. “Kernel self-representation learning based fuzzy neighborhood outlier detection.” *International Journal of Approximate Reasoning*, 198:109769, 2026. [DOI](#).

Contribution: methodology and formal analysis.

“Animal 3D Reconstruction Using Body-Shape Priors.” Chinese invention patent application No. 202611403381.7, filed Sep 2026; applicant: Sichuan University.

First-listed inventor. *Passed preliminary examination; pending publication, not granted.*

HONORS & AWARDS

First-Class Single-Category Scholarship, Sichuan University

2025 – 2026

Third Prize, Sichuan Division, Chinese Mathematics Competition for College Students

2025

Third-Class Comprehensive Scholarship, Sichuan University

2024 – 2025

Provincial Third Prize, Ascend AI Track, Huawei ICT Competition

2024

SKILLS

Methods: video diffusion, 3D Gaussian Splatting, novel-view synthesis, point-cloud processing

Programming: Python, C/C++

Frameworks: PyTorch, scikit-learn